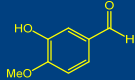




第十章 醛和酮

Aldehydes and Ketones



Vanillin
香草醛 (香兰素)



Camphor
樟脑

1

CONTENT

- 1 醛酮的结构、命名和物理性质
- 2 醛酮的制备
- 3 醛酮的反应
- 4 α, β -不饱和醛酮

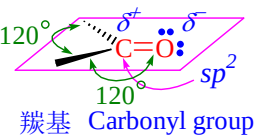
2

10.1 醛酮的结构、命名和物理性质

醛、酮的官能团是羰基，羰基碳原子为 sp^2 杂化，带部分正电荷

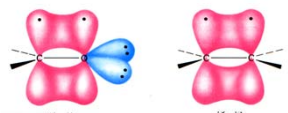
$\text{R}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{H}$
醛

$\text{R}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{R}$
酮



羰基 Carbonyl group

羰基与烯烃结构的比较



羰基



烯烃

3

10.1 醛酮的结构、命名和物理性质

醛酮的命名 (自学)

醛酮的物理性质 (自学)

4

分子极性大，与水可形成氢键。

香型：

低级醛：强烈的刺激味

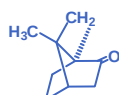
中级醛：果香味



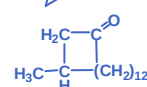
玫瑰香



香油香味主要成分



樟脑香味

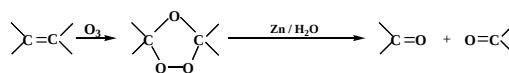


麝香香味

10.2 醛酮的制备

1. 由烯烃制备

- 臭氧解
- 氧化

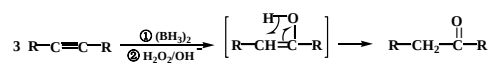
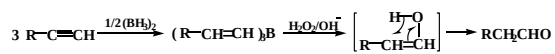
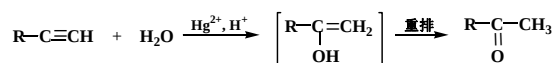


6

10.2 醛酮的制备

2. 由炔烃制备

- 水合反应
- 硼氢化—氧化反应

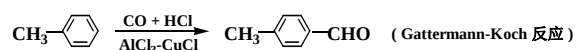
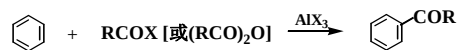


7

10.2 醛酮的制备

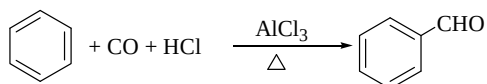
3. 由芳烃制备

- Friedel-Crafts 酰基化



8

10.2 醛酮的制备

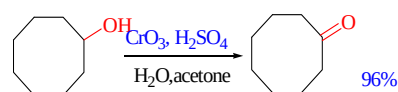
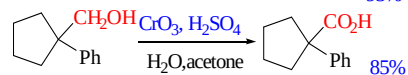
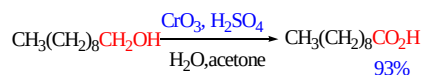


Gettman—Koch反应

9

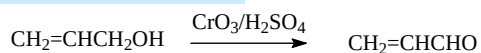
10.2 醛酮的制备

4. 由醇制备

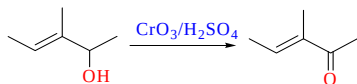
Jones' reagent — $\text{CrO}_3 / \text{H}_2\text{SO}_4$ 

• 氧化伯醇到酸，仲醇到酮，烯丙醇、苄醇到醛

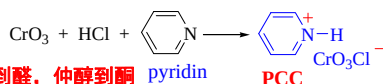
7

Johns试剂: $\text{CrO}_3 / \text{H}_2\text{SO}_4$ 

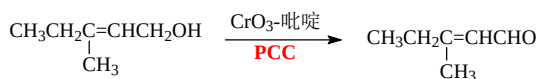
丙烯醛沸点较低，可以蒸馏得到；如不蒸馏，得到丙烯酸。



PCC (Pyridinium Chlorochromate)



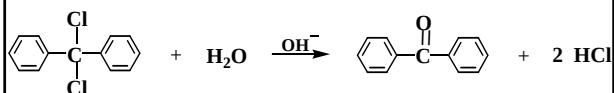
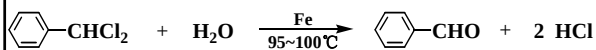
氧化伯醇到醛，仲醇到酮 pyridine PCC



11

10.2 醛酮的制备

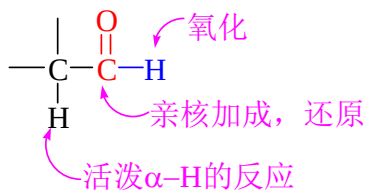
6. 从卤代烃制备——同碳二卤化物的水解



8

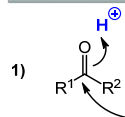
醛酮的亲核加成反应

醛酮的反应概貌

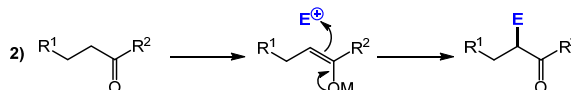


13

Reactions of Aldehydes and Ketones



Aldehydes and ketones as electrophiles



Aldehydes and ketones as nucleophiles

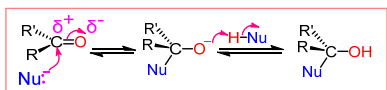
3) Oxidation and reduction reactions

结构决定性质 (反应)

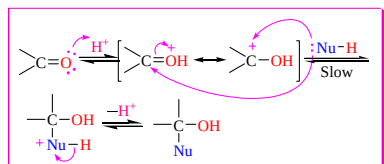
14

普遍的反应机理

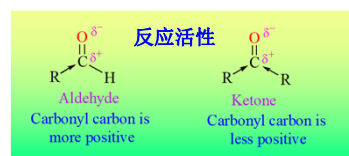
1. Addition of a strong (basic) nucleophile



2. Acid-catalyzed nucleophilic addition



15



醛 > 脂肪酮 > 芳香酮; 环酮 > 脂肪酮

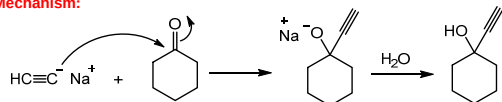
亲核试剂的分类

- 含碳的亲核试剂: RMgX , HCN ,
- 含氧的亲核试剂: H_2O , ROH
- 含硫的亲核试剂: NaHSO_3
- 含氮的亲核试剂: RNH_2 , R_2NH , $\text{H}_2\text{N-G}$

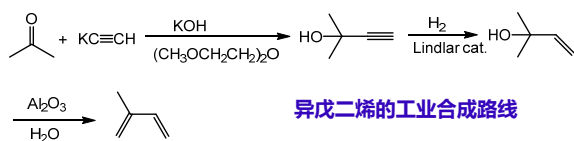
16

3. The Addition of Acetylide (炔化物)

Mechanism:



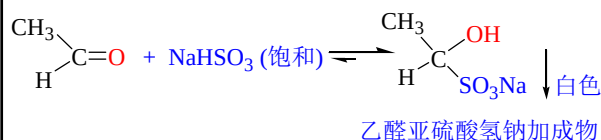
Example:



21

羰基与含硫亲核试剂的加成

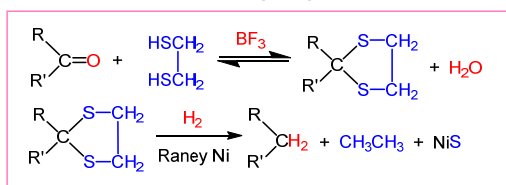
与亚硫酸氢钠的反应



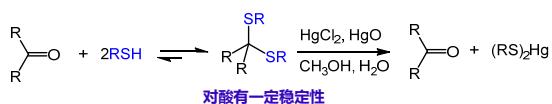
- 醛、脂肪族甲基酮、C8以下环酮能发生反应，其他酮不反应
- 本反应可用于鉴别和分离提纯

22

2. The Addition of Thiol (RSH) 强亲核性



脱除醛酮的羰基氧 (还原)

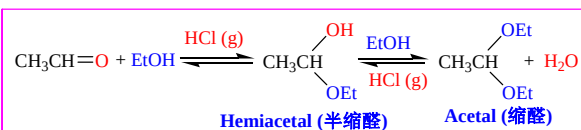


保护醛酮的羰基

23

羰基与含氧亲核试剂的加成

与醇的加成——形成半缩醛(酮)，缩醛(酮)



24

反应特点

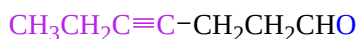
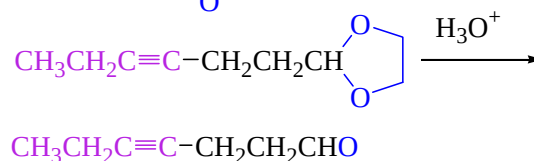
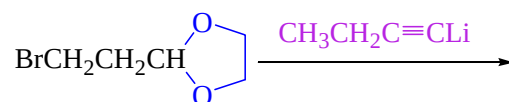
- 半缩醛(酮)无论在酸性或碱性介质中均不稳定
- 缩醛(酮)在碱性介质中稳定, 在酸性介质中不稳定
- 缩醛较易形成, 缩酮较难
- 用乙二醇形成环状缩酮反应可提高缩酮的产率

常用的催化剂:

HCl (g), TsOH / 苯, HCl / Et₂O

25

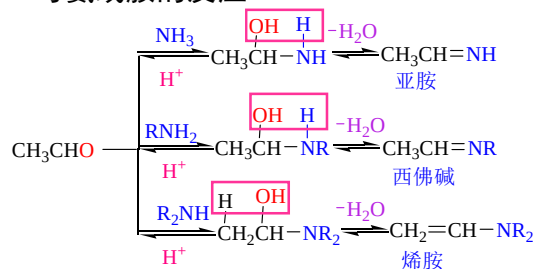
应用: 保护羰基



26

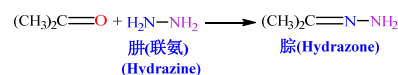
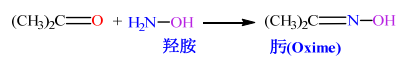
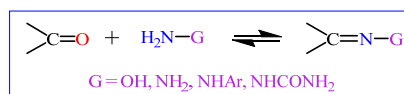
羰基与含氮亲核试剂的加成

1. 与氨或胺的反应 可逆反应, 分水操作



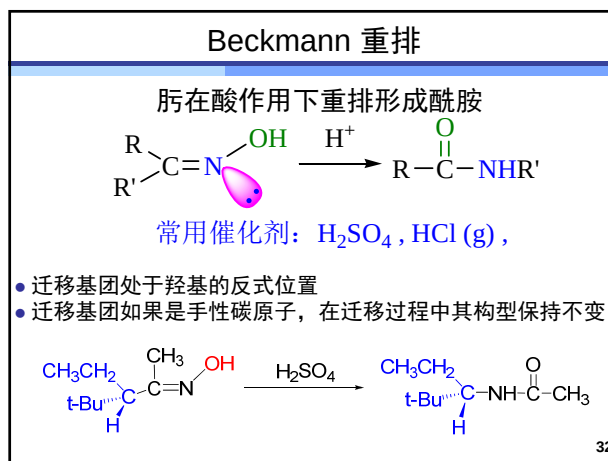
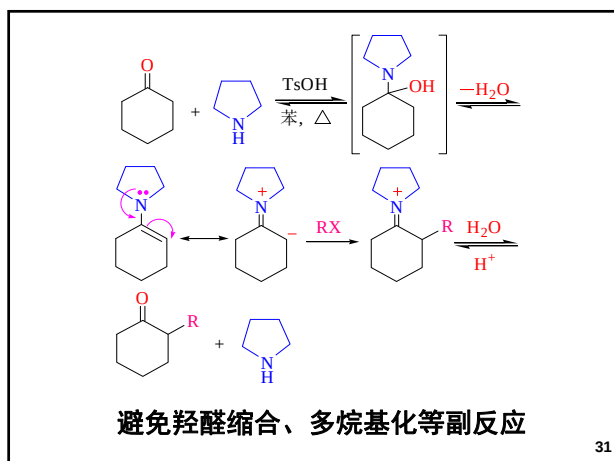
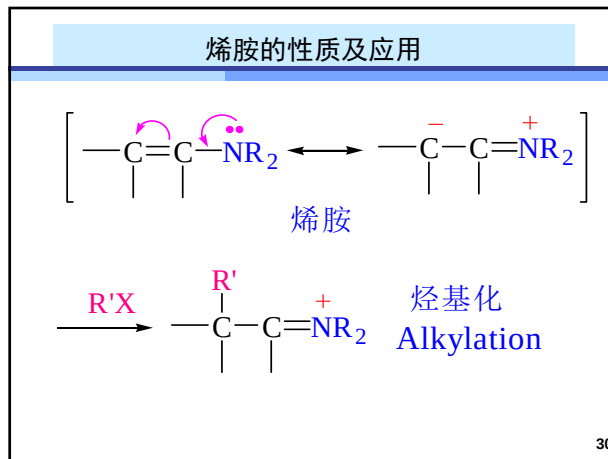
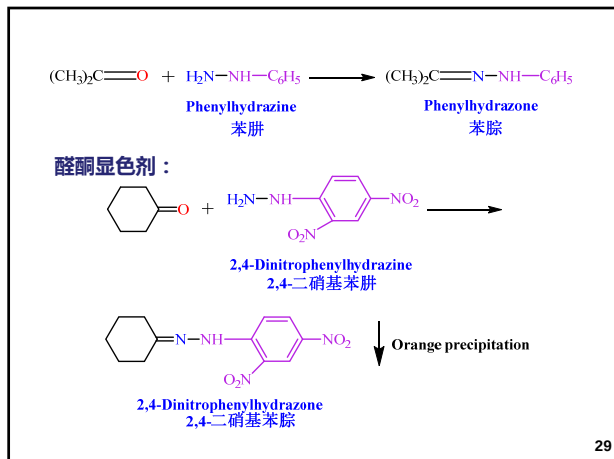
脂肪族Schiff 碱不稳定, 芳香族Schiff 碱稳定

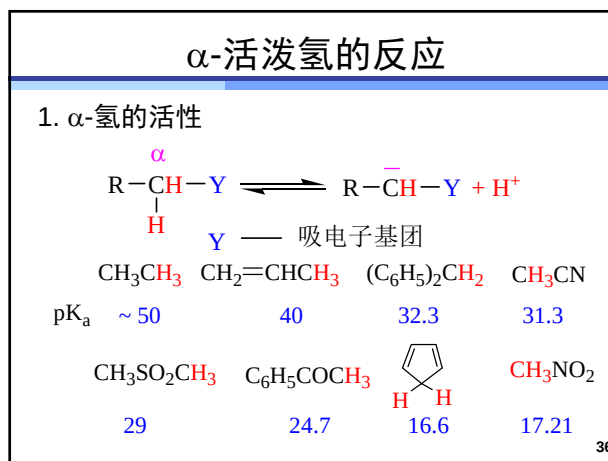
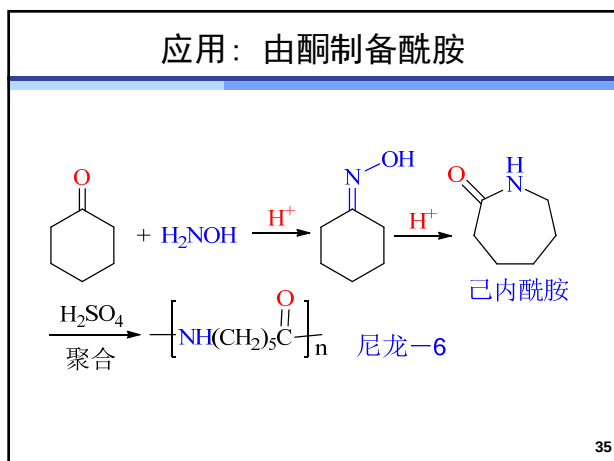
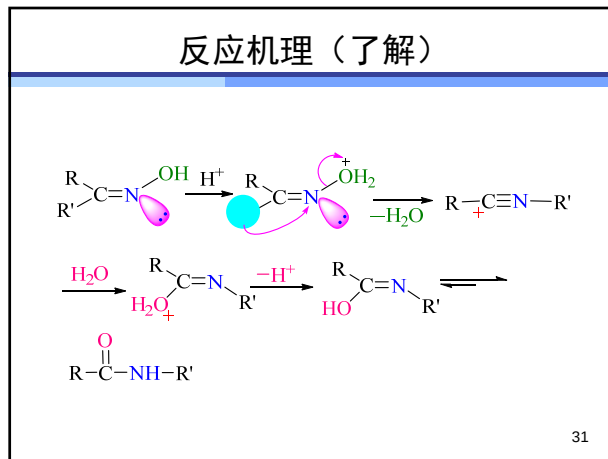
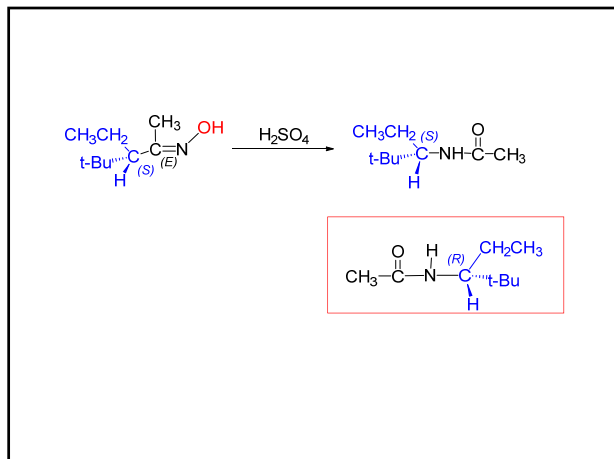
27



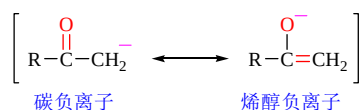
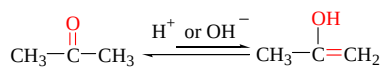
醛和非对称酮的产物构型如何? E型产物为主

28

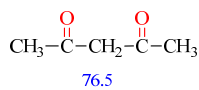
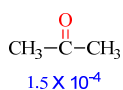




酮式和烯醇式——互变异构



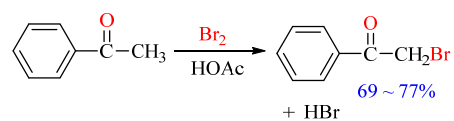
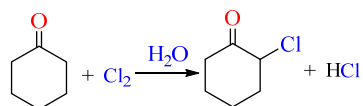
烯醇式
含量



37

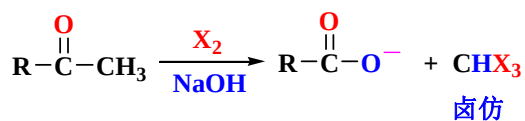
α -活泼氢的反应

2. 卤化和卤仿反应



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卤仿反应

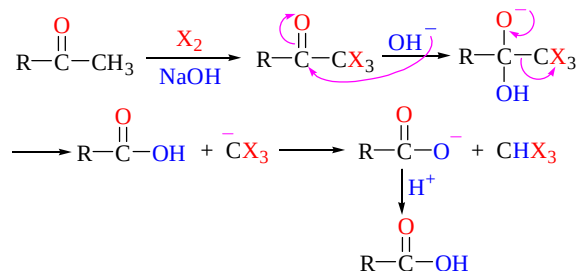


具有 $\text{—}\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{CH}_3$ $\text{—}\overset{\text{OH}}{\underset{|}{\text{CH}}}-\text{CH}_3$ 结构的
化合物都能发生反应。碘仿反应可用于鉴别。

碘仿 $\text{CHI}_3 \downarrow$ 黄色

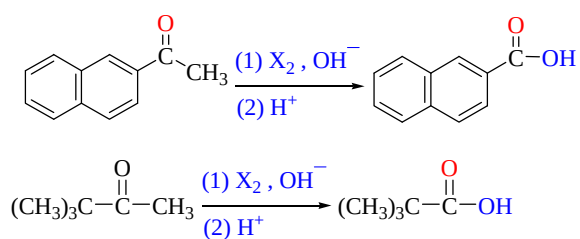
39

反应机理（了解）



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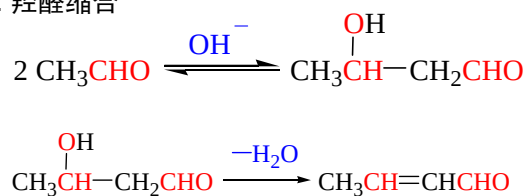
Example



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 α -活泼氢的反应

3. 羟醛缩合

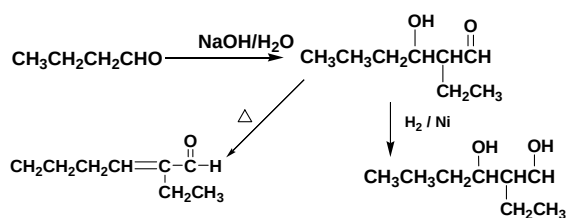


- 含有 α -H的醛(酮)在稀碱的作用下,缩合生成 β -羟基醛(酮)
- β -羟基醛(酮)易脱水形成 α, β -不饱和醛(酮)

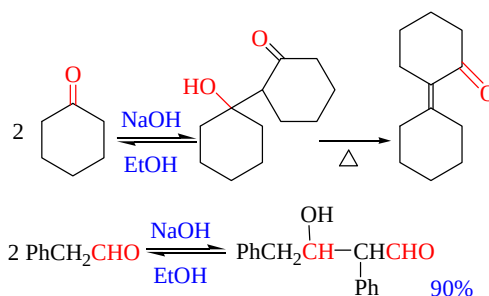
42

羟醛缩合反应的类型

(1) 自身缩合

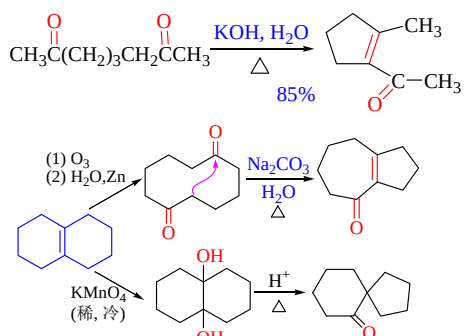


42

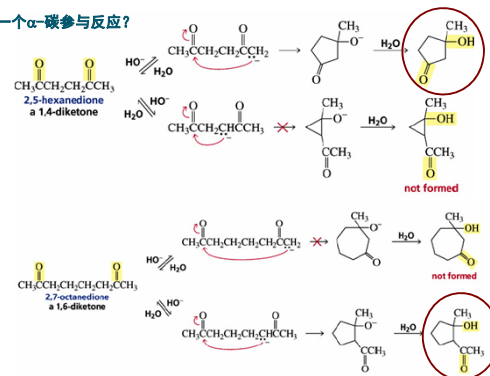


44

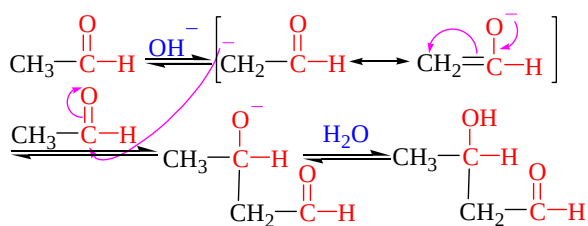
分子内缩合



45

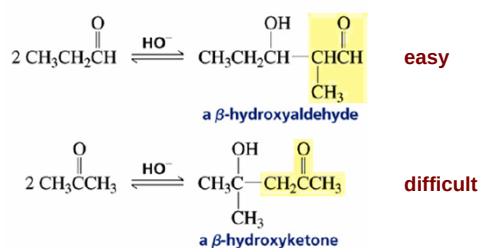
哪一个 α -碳参与反应?

反应机理 (掌握)



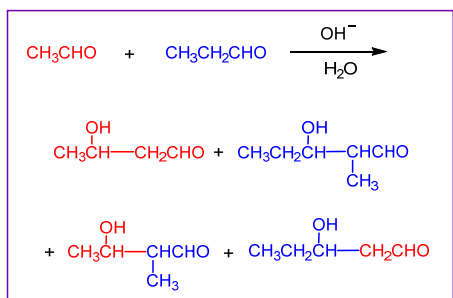
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aldol additions



羟醛缩合反应的类型

(3) 交叉缩合 Cross aldol additions/condensations



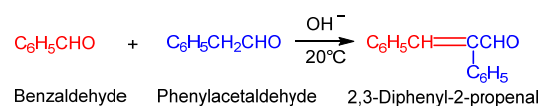
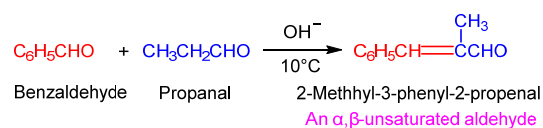
多种产物生成，包括相应的消除产物 (aldol condensation) 49

Practical Crossed Aldol Reaction

具有实用价值的羟醛缩合反应

策略一：其中一个醛无 α -H

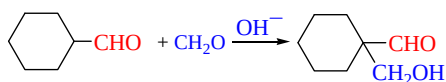
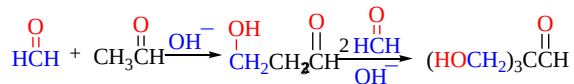
Claisen-Schmidt反应: (克莱森-施密特) 反应



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羟醛缩合反应的类型

(3) 交叉缩合 Cross aldol additions/condensations



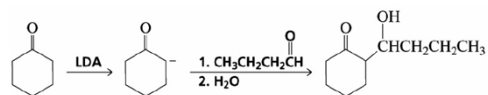
46

(3) 交叉缩合 Cross aldol additions/condensations

Practical Crossed Aldol Reaction

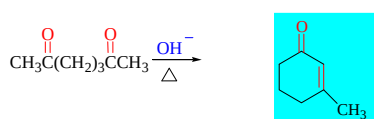
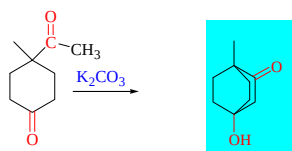
具有实用价值的羟醛缩合反应

策略二：利用酮不易发生羟醛缩合反应的性质



测试

请同学们完成下列反应

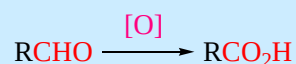


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醛、酮的氧化还原反应

1. 氧化反应

醛的氧化

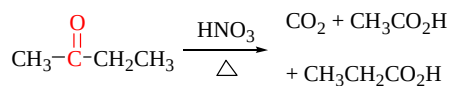
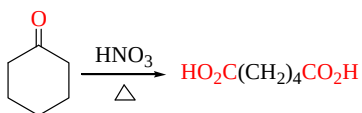


氧化剂: KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, $\text{H}_2\text{Cr}_2\text{O}_7$, H_2CrO_4 , RCO_3H , Ag_2O , H_2O_2 , $\text{Br}_2\text{-H}_2\text{O}$,

54

酮的氧化

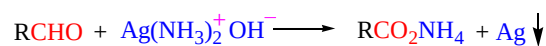
- 酮遇一般氧化剂时, 不被氧化
- 酮遇强氧化剂时, 发生碳链断裂, 氧化成酸



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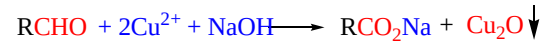
可用于鉴别醛、酮的氧化剂

1) Tollens 试剂 (银镜反应)



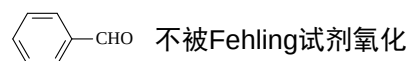
酮不发生反应

2) Fehling试剂



砖红

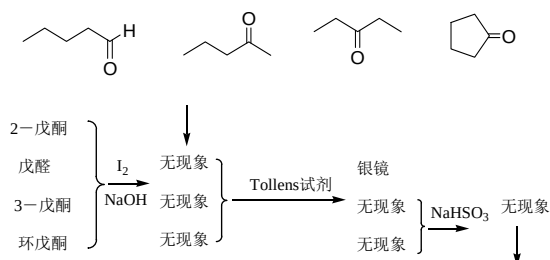
酮不发生反应



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测 试

鉴别下列化合物

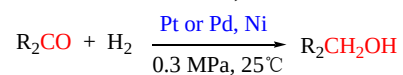
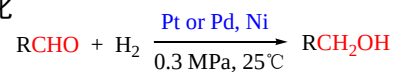


57

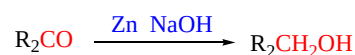
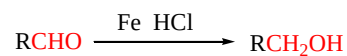
醛、酮的还原反应

2. 还原为羟基

催化氢化



金属还原剂



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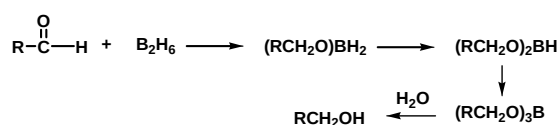
3. 用金属氢化物还原

1) $LiAlH_4$ (醚), $NaBH_4$ (醇)

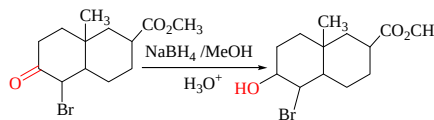
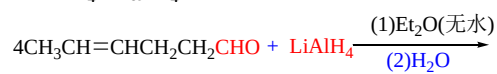
特点: 分子中的双键与叁键不被还原

(2) B_2H_6

特点: 分子中的双键与叁键可能也被还原

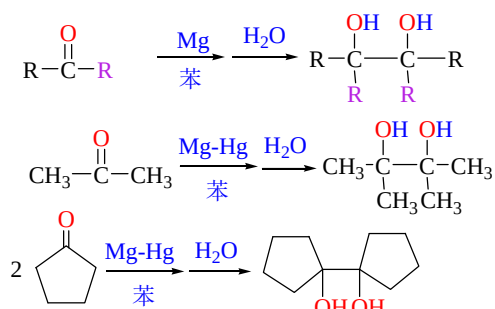


3. 用金属氢化物还原

 $LiAlH_4$ $NaBH_4$ 

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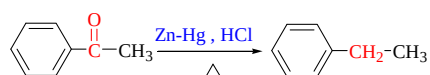
4. 双分子还原



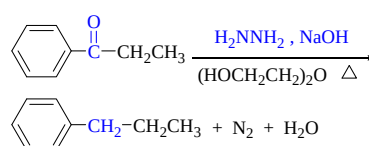
61

5. 还原为亚甲基或甲基

(1) Clemmensen Reduction

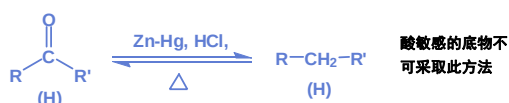


(2) Wolff-Kishner-黄鸣龙还原



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1、Clemmensen还原法

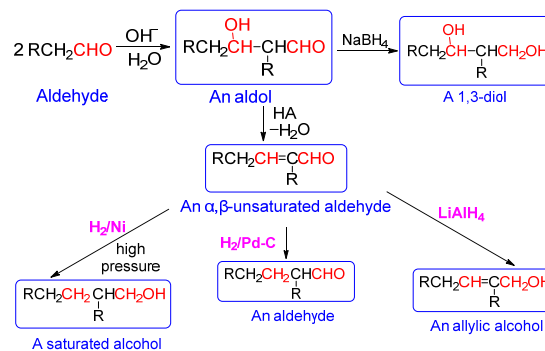


锌汞齐 (Zn-Hg)：锌粒与汞盐 (HgCl₂) 在稀HCl 溶液中
反应制得

Hg与Zn在Zn表面形成锌汞齐

还原反应在被活化的锌表面进行

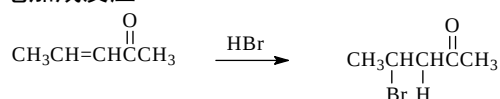
Synthetic Applications of Adol Products



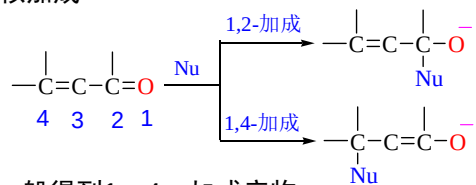
64

α, β -不饱和醛酮

1. 亲电加成反应



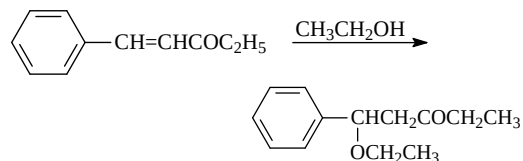
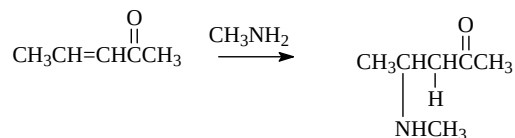
2. 亲核加成



一般得到1, 4-加成产物

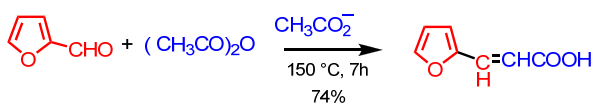
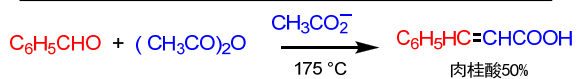
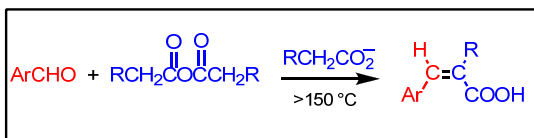
69

Example



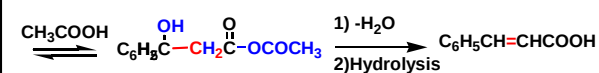
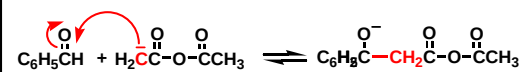
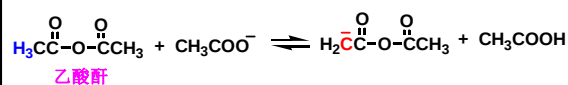
70

Perkin反应



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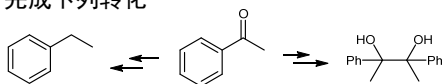
反应机理 (了解)



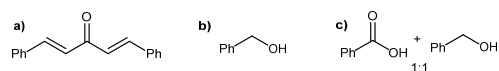
72

测 试

1, 完成下列转化



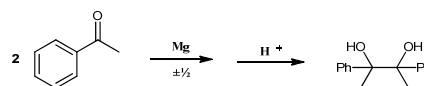
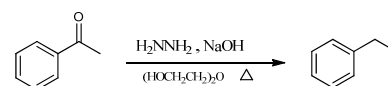
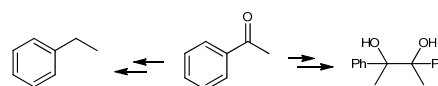
2, 以苯甲醛为原料, 利用羟醛缩合或者歧化反应, 完成下列合成



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测 试 答 案

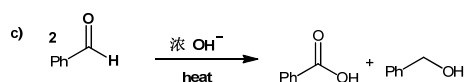
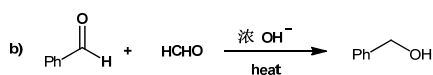
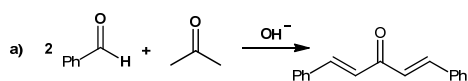
1, 完成下列转化



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测 试 答 案

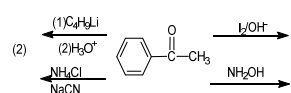
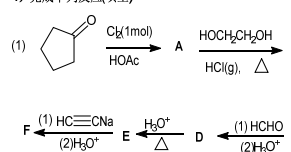
2, 以苯甲醛为原料, 利用羟醛缩合或者歧化反应, 完成下列合成



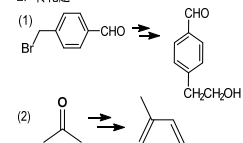
75

作业 (一)

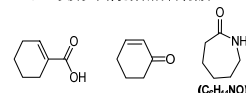
1, 完成下列反应(填空)



2, 转化题



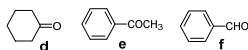
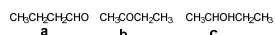
3, 以C2及以下有机物合成下列物质



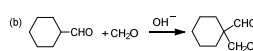
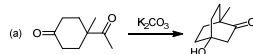
76

作业 (二)

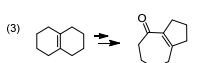
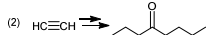
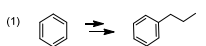
4. 用简单的化学方法鉴别下列6个化合物



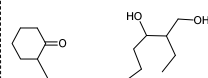
5. 写出下列反应机理



6. 转化题



7. 以C2及以下有机物合成下列物质



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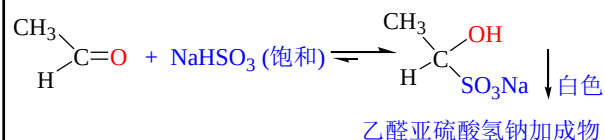
书本习题

- P198 : 10.7
 - P200: 10.8
 - P203: 10.9
 - P204: 10.10
 - P206: 10.11
- 习题
 - P214:
 - 4,5,6,7,8,9,
 - 10(1)(4).

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羰基与含硫亲核试剂的加成

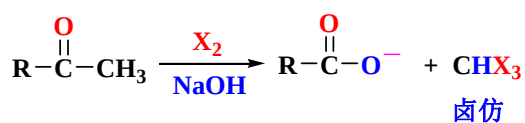
与亚硫酸氢钠的反应



- 醛、脂肪族甲基酮、C8以下环酮能发生反应，其他酮不反应
- 本反应可用于鉴别和分离提纯

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卤仿反应



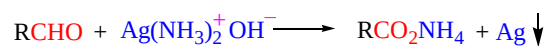
具有 $\begin{array}{c} \text{O} \\ || \\ -\text{C}-\text{CH}_3 \end{array}$ 结构的化合物都能发生反应。碘仿反应可用于鉴别。

碘仿 $\text{CHI}_3 \downarrow$ 黄色

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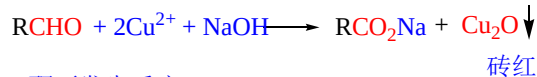
可用于鉴别醛、酮的氧化剂

1) Tollens 试剂 (银镜反应)



酮不发生反应

2) Fehling试剂



酮不发生反应

-CHO 不被Fehling试剂氧化

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